Yushi Zeng

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EDUCATION

South China University of Technology — Guangzhou, China
Doctor of Engineering in Future Technology (Advisor: Dr. Yi Cai) — Sept. 2023 - Present
Expected Graduation: June 2027

South China University of Technology — Guangzhou, China
Master of Engineering in Software Engineering (Advisor: Dr. Yi Cai) — Sept. 2020 - Jun. 2023

Jishou University — Hunan, China
Bachelor of Engineering in Software Engineering — GPA:3.47/4.00 — Sept.2016-Jun.2020

RESEARCH SUMMARY

My research focuses on knowledge transfer, continual adaptation and collaborative learning across heterogeneous models and dynamic environments. My recent work explores cloud-device test-time adaptation, class-incremental learning, and multimodal federated learning. I am particularly interested in extending these methods to reliable and efficient personalization of LLMs and intelligent agents.

RESEARCH INTERESTS: Test-Time Adaptation, Knowledge Transfer and Collaborative Learning, Multimodal Learning, Sentiment Analysis, Personalized LLMs, Agentic AI.

PUBLICATIONS

[1] **Zeng, Y.**, Ren, H., Cai, Y., Li, Y., Qin, J., & Wang, Y. (2026). Modality augmentation and task-aware dual-modal LoRAs for multi-task multimodal federated learning. *Information Processing & Management*, 63(4), 104601. **(IF 8.3, JCR Q1)**

[2] Li, Y. (Supervisor), **Zeng, Y.**, Jiang, D., Wang, Y., & Lu, G. (2026). Rethinking the knowledge gap between cloud and device models for effective co-adaptation. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(7), 9888–9900. **(IF 10.8, JCR Q1, Co-first author)**

[3] **Zeng, Y.**, Ren, H., Cai, Y., Li, Y., Wang, Y., & Li, Q. (2026). Class-incremental cloud-device collaborative adaptation with contrastive learning in dynamic changing environments. *IEEE Transactions on Neural Networks and Learning Systems*, 37(8), 3970–3984. **(IF 9.7, JCR Q1)**

[4] **Zeng, Y.**, Wang, G., Ren, H., Cai, Y., Leung, H. F., Li, Q., & Huang, Q. (2023). A knowledge-enhanced and topic-guided domain adaptation model for aspect-based sentiment analysis. *IEEE Transactions on Affective Computing*, 15(2), 709–721. **(IF 11.7, JCR Q1)**

[5] **Zeng, Y.**, Wang, G., Ren, H., & Cai, Y. (2022, June). Enhance cross-domain aspect-based sentiment analysis by incorporating commonsense relational structure (Student Abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 36, No. 11, pp. 13105–13106). **(CCF-A)**

[6] Ren, H., **Zeng, Y.**, Cai, Y., Ye, Z., Yuan, L., & Zhu, P. (2024, May). Grounded multimodal procedural entity recognition for procedural documents: A new dataset and baseline. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (pp. 7971–7981). **(CCF-B; Co-first author)**

[7] Ren, H., **Zeng, Y.**, Cai, Y., Zhou, B., & Lian, Z. (2023, July). Constructing procedural graphs with multiple dependency relations: A new dataset and baseline. In *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 8474–8486). **(CCF-A; Co-first author)**

[8] Zhou, B., Ren, H., Cai, Y., Lian, Z., Zhu, P., **Zeng, Y.**. MPPQA: Structure-Aware Extractive Multi-span Question Answering for Procedural Documents. *Natural Language Processing and Chinese Computing. (NLPCC 2024.)*, vol 15359. **(CCF-B)**

[9] H. Ren, Y. Cai, **Y. Zeng**, J. Ye, H. -f. Leung and Q. Li, (2022) "Aspect-Opinion Correlation Aware and Knowledge-Expansion Few Shot Cross-Domain Sentiment Classification," in *IEEE Transactions on Affective Computing*. vol. 13, no. 4, pp. 1691-1704 **(IF 11.7, JCR Q1)**

[10] Ren, H., Cai, Y., & **Zeng, Y.**. (2022). Aspect-Opinion Sentiment Alignment for Cross-Domain Sentiment Analysis (Student Abstract). *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(11), **(CCF-A)**

RESEARCH EXPERIENCE

Research on Cloud-Device Collaborative Learning — PengCheng Laboratory
Core Student Researcher — Sept 2024 — Present

- **Knowledge-Gap-Aware Cloud-Device Adaptation (IEEE TCSVT, 2026):** Developed information-aware sample selection and knowledge-gap-guided distillation strategies to enable efficient knowledge transfer between cloud and device models under distribution shifts.
- **Class-Incremental Collaborative Learning (IEEE TNNLS, 2026):** Proposed a contrastive prototype-based collaborative learning framework to support class-incremental adaptation and improve generalization in dynamically changing environments.
- **Reliability-Aware Multi-Expert Co-Adaptation (ACM TOMM, Under Review):** Proposed a multi-expert cloud-device adaptation framework with reliability-aware expert routing, collaborative target construction, and student-centric distillation to enable reliable knowledge transfer from complementary cloud experts.

- **Dynamic Heterogeneous Multi-Device TTA (Manuscript in Preparation; Target Venue: CVPR 2027):** Investigating collaborative adaptation across heterogeneous devices experiencing distinct, evolving, and recurring domain shifts. Exploring domain-aware memory and reliability-aware knowledge sharing to enable continual adaptation while mitigating negative transfer.

Research on Multimodal Federated Learning PengCheng Laboratory
Core Student Researcher Sept 2023 — Sept 2024

- **Multi-Task Multimodal Federated Learning (IP&M, 2026):** Developed a modality augmentation and task-aware dual-modal LoRA framework to address missing modalities, mitigate modality bias, and facilitate knowledge transfer across heterogeneous tasks and distributed clients.

Research on Standard knowledge graph construction and application Jingshan WPS
Research Member Sept 2020 — Nov 2021

- **Knowledge Graph Construction**: The Schema construction for the standard domain and the knowledge extraction Extract the entity-relation triples by utilizing the inherent document structure features of the standard documents. Moreover, the BILSTM-CRF and BERT-CRF models are used to extract the fine-grained entities. Then, the extracted standard knowledge triples are stored in the neo4j database and we conduct the visualization analysis.
- **Standard Knowledge Recommendation**: We adopt a content-based recommendation method which uses the user-entered keywords to retrieve and link the relevant entities from standard knowledge graph.
- Standard conflict detection: We recognize the candidate entity from the standard drafts and kink them from the knowledge graph and identify the suspected conflicting entries through the text comparison.

INTERNSHIP EXPERIENCE

Huawei | "AI Applications in Wireless Information Experience" Shenzhen, China
IT Platform Department 3 Jun. 2022 — Sep. 2022

- Worked on an internal Huawei project involving heterogeneous knowledge bases, solving cross-graph retrieval challenges by linking product and terminology knowledge graphs to improve the relevance of technical entity retrieval.
- Built an entity recognition, linking, and disambiguation pipeline with Jieba, pre-trained BERT embeddings, and cosine similarity, reducing ambiguity in product names and attribute descriptions and improving matching accuracy.
- Improved system deployment and API performance by designing a multi-threaded access strategy, raising entity linking accuracy to over 90% and ensuring 99% of response times remained under 200 ms.

TEACHING EXPERIENCE

South China University of Technology Guangzhou, China
Graduate Teaching Assistant, Computer Architecture [Fall 2023]

- Graded homework assignments, held regular office hours for student Q&A.
- Assisted the instructor in supervising and guiding lab sessions for computer architecture experiments.

AWARDS & COMPETITION

- Outstanding Student, 2025 Tencent Rhino-Bird Open Source Talent Development Program 2025
- Wuliangye Outstanding Student Scholarship 2023
- China National Scholarship 2022
- First-Class Postgraduate Scholarship 2022
- AAAI Student Scholarship Program 2022
- Certificate of Completion, Summer Research Experience Programme for Mainland and Taiwan Students, The Chinese University of Hong Kong 2022
- Outstanding Graduate, Jishou University 2020
- First-Class and Second-Class Scholarships, Jishou University (Awarded over Four Consecutive Years) 2017–2020
- Third Prize, The 10th Lanqiao Cup National Software and Information Technology Competition 2019

SKILLS & INTERESTS

- **Languages:** Mandarin (Native), Cantonese (Fluent), English (Fluent,CET6)
- **Programming Languages:** Python (PyTorch), JAVA, C
- **Software and Tools:** Microsoft Office (Word, Excel, and PowerPoint), Zotero, LaTex, Markdown
- **Interests:** Badminton, Fitness